

MP204 – Electricity & Magnetism

Problem Set 6

Due in the assignment box by 5pm, Friday 8 May 2015

1. Consider an electromagnetic (EM) wave travelling through empty space described by the electric and magnetic fields

$$\vec{E} = 3E_0 \cos(z - ct) \hat{i}, \quad \vec{B} = \vec{B}_0 \cos(z - ct),$$

where E_0 is a positive constant and $\vec{B}_0$ is a constant vector.

- (a) In which direction is this wave travelling?
(b) Find the magnitude (in terms of E_0) and direction of $\vec{B}_0$.
2. We have a standing EM wave whose electric field component is given by

$$\vec{E} = E_0 \cos(\kappa y) \cos(\omega t) \hat{k}$$

where E_0 is the field's magnitude, $\kappa = 2\pi/\lambda$ its wave number and $\omega = 2\pi\nu$ its angular frequency, with $\lambda\nu = c$.

- (a) Verify that this satisfies the wave equation.
(b) Using Maxwell's Third equation

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t},$$

find the magnetic field component of this wave.

- (c) Using whatever trigonometric identity seems appropriate, find the functions f_+ and f_- such that $\vec{E}$ may be written in the form

$$\vec{E} = [f_+(y + ct) + f_-(y - ct)] \hat{k}.$$

3. (a) Compute the energy density u and Poynting vector $\vec{S}$ for the propagating EM wave in Problem 1.
(b) Now do the same for the standing wave in Problem 2.
4. Suppose we have an electromagnetic field not in a vacuum, but in a medium with conductivity σ ; this means that the charge density ρ is still zero, but the current density is given by Ohm's law $\vec{J} = \sigma \vec{E}$. Using this, Maxwell's equations and some vector calculus (like what we did in lecture), show that the electric and magnetic fields satisfy the equations

$$\begin{aligned} \nabla^2 \vec{E} &= \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} + \mu_0 \sigma \frac{\partial \vec{E}}{\partial t}, \\ \nabla^2 \vec{B} &= \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} + \mu_0 \sigma \frac{\partial \vec{B}}{\partial t} \end{aligned}$$

(Remember that $\vec{\nabla} \times (\vec{\nabla} \times \vec{V}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{V}) - \nabla^2 \vec{V}$ for any vector field $\vec{v}$.)

Remark: You can solve these equations (although I'm not asking you to) to find that the solutions have the general form $e^{-x/L} \sin(\kappa x - \omega t)$, where L is a quantity which depends on both σ and ω . The exponential in front means that the wave becomes *weaker* as it travels through the material, a phenomenon known as *attenuation*. L is thus a measure of how far a wave can penetrate into the medium before it dies away, the "attenuation length". As σ approaches zero, the medium looks more like a vacuum, and L gets larger and larger; it travels further before dying away. But if the material is highly conductive, L is quite small; the wave can barely penetrate the material at all. A perfect conductor has $\sigma \rightarrow \infty$ and thus $L = 0$, which means $\vec{E} = \vec{0}$, exactly what we expect.

VECTOR CALCULUS FORMULAE IN CARTESIAN COORDINATES

Cartesian coordinates (x, y, z) with constant unit direction vectors $\hat{i}, \hat{j}, \hat{k}$

- gradient of a function $f(x, y, z)$:

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

- divergence of a vector $\vec{A}(x, y, z) = A_x(x, y, z)\hat{i} + A_y(x, y, z)\hat{j} + A_z(x, y, z)\hat{k}$:

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

- curl of a vector $\vec{A}(x, y, z) = A_x(x, y, z)\hat{i} + A_y(x, y, z)\hat{j} + A_z(x, y, z)\hat{k}$:

$$\vec{\nabla} \times \vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{k}$$

- Laplacian of a function $f(x, y, z)$:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$